

Your thesis title

Your name

DEPARTMENT OF PHYSICS AND ASTRONOMY, BATES COLLEGE, LEWISTON, ME 04240

Your thesis title

A Senior Thesis

Presented to the Department of Physics and Astronomy

Bates College

in partial fulfillment of the requirements for the

Degree of Bachelor of Arts

by

Your name

Lewiston, Maine

December 16, 2008

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Acknowledgments

Mom, Dad, your roommate, maybe even your thesis advisor – give credit where credit’s due! For my part, I’d like to thank the Math department for making such a nice L^AT_EX template that we could use as a base for our Physics template.

Introduction

You can, if you wish, write an introduction.

CHAPTER 1

Some Things you can do in L^AT_EX

1. Writing text in L^AT_EX

You can write text in L^AT_EX as you would in any other word processing or typesetting software. You can incorporate *different varieties of typeface*. What's really exciting about writing text in L^AT_EX is that it's very easy to incorporate weird symbols, like α or $\otimes$ or ∞ into your text. Note that the weird symbols appear in between \$ signs – this means that they occur in *math mode*.

You create a new paragraph just by putting in a blank line. Leaving extra space or extra lines in between paragraphs won't change their spacing at all. L^AT_EX will automatically indent your paragraphs, so you don't have to worry about indenting. You can, however prevent it from indenting your paragraphs if you'd like. But that's getting a little ahead of ourselves.

2. Writing equations in L^AT_EX

It is beautifully easy to write equations, whether they occur inline with the text $\Psi(x, t) = Ae^{i(kx - \omega t)}$ or as a separate line

$$(1.1) \quad \Psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-iE_n t/\hbar}.$$

To continue a paragraph after an equation, make sure that there's no blank line between the equation and your next line of text. Equations that are in a separate line will be numbered, and it's easy to refer to Eq. 1.2 by labelling your equations

$$(1.2) \quad i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \hat{H} \Psi(x, t).$$

Are you getting tired of writing `begin{eqnarray}` and `end{eqnarray}` yet?¹ You can simplify your life by defining abbreviated symbols for commands:

$$(1.3) \quad \langle x | \Psi(t) \rangle = \Psi(x, t).$$

For simplicity, I'd suggest putting all of your new commands in a separate include file.

When writing a derivation with multiple steps, it is often useful to use an equation array

$$\begin{aligned} \int_0^{2\pi} \sin^2 x dx &= \int_0^{2\pi} \frac{1}{2} (1 - \cos 2x) dx \\ &= \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) \Big|_0^{2\pi} \\ &= \pi. \end{aligned}$$

¹Notice how I made a } by putting a backslash in front of the symbol. The curly brace has special meaning in L^AT_EX, so if you want to make one, you have to let the program know that you want the symbol.

Note that you can suppress equation numbering, which is often desirable in derivations!

The syntax for equation arrays is similar to the syntax for arrays within equations. For example,

$$(1.4) \quad \hat{H} = \begin{pmatrix} \Delta & \Omega & 0 \\ \Omega^* & 0 & \kappa \\ 0 & \kappa^* & -\Delta \end{pmatrix}$$

There are many more complicated things you can do with equations, but this should be enough to get started!

3. Tables, lists, etc... etc...

If you want to make a table, you'll need to specify the justification for each of the columns; $\text{\TeX}$ will automatically calculate the column widths for you.

Name	Age	Town	Time
Joe Schmoe	23	Rumford	1:04
Mary Q. Contrary	44	Yarmouth	1:12
Baxter Boulevard	32	Portland	1:42
Lass Plaiice	60	Lewiston	1:57

TABLE 1. Not too many people showed up for the first, and probably last, annual Androscoggin swimming race.

You might also, for example, want to

- (1) make lists
- (2) number your lists
 - use bullet points
 - find better items to put in your bullet points

4. Figures

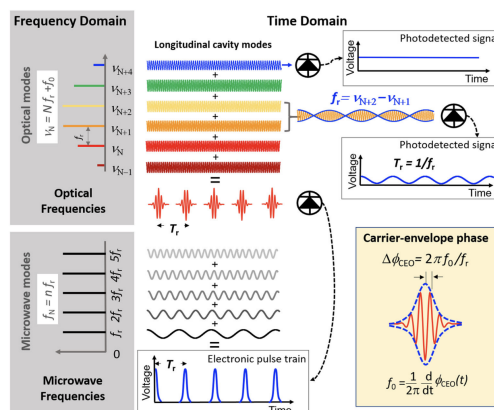


FIGURE 1.1. This is a very bad figure.

5. References

test citation [1]

CHAPTER 2

Some More Things you can do in $\text{\LaTeX}$

WOOOOO

Appendix A

Long derivations, marginally relevant details, and the like can be relegated to your appendix, which unlike the mammalian counterpart will never require removal.

Bibliography

- [1] D.E. Chang et al. “Quantum optics with surface plasmons”. In: *Physical Review Letters* 97 (2006), p. 053002.